

Nicolas Velasquez Lopez

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PROFESSIONAL SUMMARY

Mechatronics Engineer and Solutions Engineer working across applied artificial intelligence, cloud architecture, automation, and robotics. Experienced in leading technical discovery, translating business needs into architectures, demos, and proofs of concept, and collaborating with sales, product, and engineering teams to deliver practical systems. Combines customer-facing communication with hands-on work in Python, LLM and agentic workflows, cloud platforms, APIs, electronics, and robotic control.

PROFESSIONAL EXPERIENCE

Solutions Engineer — Truora · Bogotá, Colombia · Oct 2025–present

- Leads technical discovery with customers to understand architecture, workflows, requirements, business objectives, and success criteria.
- Designs and presents tailored demos and proofs of concept with sales, product, and engineering teams.
- Builds and operates LLM and agentic workflows for pre-sales and pilot operations, including tool use, orchestration, and process automation.
- Uses HubSpot for pipeline and ticket follow-up and built an automation to streamline the ticket-tracking workflow.

Independent AI & Automation Consultant — UnMecaNiko · Bogotá, Colombia · Jul–Sep 2025

- Built multi-agent assistants, document automation, and production integrations using Python, n8n, APIs, cloud services, and Supabase.
- Developed a public technical portfolio documenting deployed projects across AI, cloud, robotics, and hardware.

Named Accounts Cloud Solutions Architect — Huawei · Bogotá, Colombia · Jul 2024–Jul 2025

- Led technical discovery and end-to-end cloud solution design for named enterprise accounts, balancing performance, availability, cost, scalability, and security.
- Architected a cloud migration for a major Colombian technology company in the betting industry.
- Delivered technical training and webinars on AI and Kubernetes for technical and business audiences in Colombia and Ecuador.

Cloud Solutions Architect Intern — Huawei · Bogotá, Colombia · Feb–Jul 2024

- Supported market and competitive analysis, customer communication, technical proposals, and solution improvements.

Scientific Instructor — Colombian Astronautical Association, Space Call · Nov 2023–Feb 2024

- Mentored students in programming, electronics, mechanics, and Morse code to build a board integrated into the RASCube-1 educational CubeSat, launched in 2025 as part of a seven-country project.

Co-founder, Coordinator & Instructor — Lápiz Labs · Jul 2023–Feb 2024

- Co-founded and operated a robotics and programming education initiative serving 40+ students through approximately 80 classes with five instructors.
- Designed curricula, trained instructors, coordinated operations, and prepared students for robotics competitions.

SELECTED PROJECTS

Agent-assisted pre-sales operating system — Designed a repeatable system for customer discovery, research, demos, and pilot operations using structured context and AI agents. [Public walkthrough](#)

DomiChat — Built a multi-agent WhatsApp assistant for logistics using Gemini, n8n, Supabase, and production integrations. domichat.com.co

CNC Magia Roja — Designed and integrated electronics, GRBL firmware, stepper motors, sensors, wiring, and safety systems for a functional CNC used in a family business.

TurtleBot trajectory control — Implemented and validated differential-drive trajectory control using ROS 1, Gazebo, MATLAB/Simulink, Lyapunov control, and odometry in simulation and hardware.

CORE SKILLS

Solutions engineering: technical discovery, solution architecture, demos, proofs of concept, technical proposals, customer enablement.

Applied AI & automation: Python, LLM applications, AI agents, RAG, orchestration, n8n, MCP, APIs, Supabase.

Cloud & infrastructure: cloud architecture, Huawei Cloud, Kubernetes, Docker, Linux, Vercel, security, scalability, availability, cost optimization.

Mechatronics & robotics: electronics integration, sensors, microcontrollers, stepper motors, GRBL, ROS 1, Gazebo, MATLAB/Simulink, control systems; currently learning ROS 2.

EDUCATION & CERTIFICATIONS

- **Mechatronics Engineering** — Universidad Militar Nueva Granada, 2019–2023.
- **HCIP Cloud Service Solutions Architect** — Huawei, Mar 2025.
- **HCIA Artificial Intelligence** — Huawei, May 2024.
- **HCIA Cloud Service** — Huawei, Apr 2024.
- **MCP** — Platzi, 2025.
- **Professional Scrum Course** — Scrum.org.
- **English** — Advanced, CEFR C1 — EF.

SELECTED ACHIEVEMENTS

- Senior champion, UMNG Programming Olympics (2021).
- Robotics category champion, UMNG Engineering Week (2022).
- Second place, Innovation category, Latin American Innovation Rally (2023).
- Judge, World Robot Olympiad Colombia (2023).
- Scientific instructor, RASCube-1 educational CubeSat project.